TEAM VARUNA

ORION-Hackathon 2K26

Ship-to-Shore and Island delivery drone

Hackathon Report

Submitted By

Ishaan Mundhra
Ishaan Shankar
Yash Gupta
Pusala Prashanth

Indian Institue of Space Science and Technology

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Abstract

Autonomous littoral resupply in contested maritime environments demands rapid, runway-independent logistics capable of operating in severe wind conditions and executing dynamic platform landings. This report details the system architecture, aerodynamic sizing, and simulation of a hybrid QuadPlane Vertical Take-Off and Landing (VTOL) unmanned aerial vehicle engineered for a 15 km ship-to-shore supply mission. Tasked with delivering a 5 kg payload against 12 m/s coastal winds, the platform leverages a 12-foot wingspan composite airframe protected by marine-grade IP67 sealing and anti-corrosion coatings. To optimize flight endurance, the propulsion and power architecture is segregated: four 25 cm lift rotors driven by a high-discharge 12S LiPo pack manage the 4000 W instantaneous hover transients, while a single 50 cm pusher propeller powered by a custom 12S4P Li-ion pack ensures high-efficiency 24 m/s cruise flight. The decentralized avionics stack integrates a Pixhawk 6X flight controller with a Raspberry Pi 5 edge computer. To minimize power draw and processing overhead during the 30 km round trip, the aircraft relies on optical flow odometry for low-compute transit navigation, activating a YOLOv8- driven vision system strictly for precision terminal intercepts on platforms moving at up to 2 m/s. With a tightly balanced Maximum Takeoff Weight (MTOW) of 26.9 kg and an onboard energy capacity exceeding 1180 Wh, the proposed architecture is designed to satisfy the payload, failsafe, and 30% abort-reserve requirements, subject to detailed energy-budget and flight validation.

Nomenclature

AR	Aspect ratio of the wing
b	Wingspan (m)
c_{root}	Root chord length (m)
c_{tip}	Tip chord length (m)
$\bar{c}$	Mean geometric chord length (m)
C_L	Lift coefficient
$C_{L,\max}$	Maximum lift coefficient
$C_{L,\alpha}$	Lift-curve slope (rad^{-1})
C_D	Drag coefficient
C_{D0}	Zero-lift drag coefficient
D_{rotor}	Lift rotor diameter (m)
e	Oswald efficiency factor
E	Battery energy capacity (Wh)
h	Wing height parameter (m)
I	Electrical current (A)
K	Induced drag factor
L_f	Fuselage length (m)
L/D	Lift-to-drag ratio
MAC	Mean aerodynamic chord (m)
$MTOW$	Maximum take-off weight (kg)
P	Electrical power (W)
Re	Reynolds number
S	Wing planform area (m^2)
t/c	Airfoil thickness-to-chord ratio
T/W	Thrust-to-weight ratio
V	Airspeed (m/s)
V_{cruise}	Cruise velocity (m/s)
V_{stall}	Stall velocity (m/s)
V_{avail}	Available payload & battery vol (m^3)
W/S	Wing loading (N/m^2)
W_f	Fuselage width (m)
H_f	Fuselage height (m)
α	Angle of attack (°)
γ	Flight-path angle (°)
θ	Wing taper angle (°)
λ	Wing taper ratio
ρ	Air density (kg/m^3)

Abbreviations

VTOL	Vertical Take-Off and Landing
UAV	Unmanned Aerial Vehicle
ESC	Electronic Speed Controller
LiPo	Lithium Polymer Battery
Li-ion	Lithium-ion Battery
PX4	PX4 Autopilot Flight Control Software
ROS 2	Robot Operating System 2
SITL	Software-In-The-Loop
VIO	Visual-Inertial Odometry
IP67	Ingress Protection Rating 67

1 Introduction

1.1 Background

Modern defense technology has advanced far beyond the latest ammunition and firearms. We are now in the age of unmanned autonomous systems that fight our battles far beyond the frontier. Another advantage they could offer in wartime situations is the transport of critical equipment or provisions to soldiers stationed in remote environments, such as a deserted island in enemy territory. A naval vessel far from the island ($\sim 15km$), could deploy a VTOL drone, equipped to carry the provisions (payload $\sim 5kg$), facing wind gusts up to $12m/s$.

1.2 Objectives

- Landing or dropping the payload onto a moving platform (presumably the ship being rocked by the waves).
- Using corrosion-resistant materials to make up the exterior of the vehicle and use sealants to keep the internals protected.
- Resist the wind gusts up to $12m/s$, while precisely landing on the ship's deck which is moving by $2m/s$ horizontally.
- Perform precision landing utilizing either a vision system or a beacon.
- Maintain an energy reserve for the return flight and incorporate lost-link failsafes.

2 Literature Review

2.1 Existing Work

Hybrid UAV Design and Aerodynamic Evaluation[1]

Gu et al. (2017) presented the design and experimental verification of a hybrid vertical take-off and landing (VTOL) unmanned aerial vehicle. Their work covered aerodynamic analysis, mechanical design, onboard system integration and flight-controller development. The authors experimentally evaluated the aircraft's aerodynamic performance, stability, endurance and range, and demonstrated autonomous flight in an outdoor environment. The study highlights the importance of treating aerodynamic design as part of an integrated aircraft development process rather than considering wing geometry in isolation. For the present wing-design study, this work provides useful methodological reference for evaluating aerodynamic performance in relation to the intended aircraft's operating requirements. However, the paper's main focus is the overall UAV configuration and its experimental validation, rather than a systematic comparison of individual airfoils. Therefore, the selection of NACA 4415 in the present study is based on its own aerodynamic analysis and suitability for the specified design conditions.

Design and Aerodynamic Considerations for Hybrid UAVs[2]

Rehan et al. (2022) presented a comprehensive review of (VTOL) vertical take-off and landing hybrid unmanned aerial vehicles, examining their design approaches, aerodynamic characteristics, dynamic modelling and control strategies. Hybrid VTOL UAVs combine the vertical take-off and landing capabilities of multirotor aircraft with the efficient forward-flight characteristics of fixed-wing aircraft. The authors discussed the importance of selecting an appropriate aircraft configuration and integrating aerodynamic design with stability, propulsion and flight-control requirements. The review highlights that the aerodynamic performance of the fixed wing plays an important role in determining forward-flight efficiency, endurance and operational range. It also emphasises the need to consider the interaction between the aircraft's aerodynamic characteristics and its dynamic behaviour when developing a reliable UAV. This work provides a useful foundation for the present wing-design study, where airfoil selection is an important step towards achieving the required lift and drag characteristics. NACA 4415 was selected for further evaluation because of its moderate camber and 15% thickness, which make it a reasonable candidate for investigating lift generation, aerodynamic efficiency and structural integration. The present study extends this design approach by evaluating the airfoil's performance under the specified operating conditions using Foil.tools. However, since Rehan et al. focus on hybrid UAV configurations rather than a direct comparison of individual airfoils, the paper supports the broader design methodology rather than establishing the superiority of NACA 4415.

3 Methodology

3.1 Proposed Approach

We shall now discuss the approaches that we have adopted to solve the various problems:

Architecture Evaluation: Multirotor v/s Hybrid VTOL

Multirotors excel in precision maneuvering, offering highly stable hover capabilities and a compact physical footprint. In a naval environment, they are adept at matching the deck motion of a vessel (up to 2 m/s) or executing precise winch operations in confined drop zones. However, the fundamental flaw of the multirotor architecture in extended resupply missions is its aerodynamic inefficiency. Because multirotors rely entirely on their lift motors to remain airborne, they consume energy continuously at an exorbitant rate. Attempting to push a multirotor carrying a 5 kg pay-

load across a 30 km round trip against constant 12 m/s coastal winds demands a prohibitive battery mass, leaving minimal margin for safety or lost-link failsafes.



Figure 1: Multirotor

A hybrid VTOL platform eliminates the need to choose between launch flexibility and cruise efficiency. By incorporating four dedicated lift rotors, the aircraft can take off vertically from a crowded ship deck or an unprepared island clearing without needing a runway or catapult. Once airborne, it transitions to forward flight, utilizing a dedicated pusher propeller and its 12-foot wingspan to generate lift aerodynamically rather than purely through downward thrust. Fixed-wing cruise flight uses significantly less power per kilometer compared to a multirotor, drastically extending endurance and operational range.



Figure 2: Hybrid VTOL

3.2 System Design

The proposed platform operates on a Quad-Plane hybrid VTOL architecture, integrating a 12-foot wingspan fixed-wing airframe with a dual-boom quad-rotor system. This structural separation isolates the thrust vectors into two distinct groups: four 25 cm propellers dedicated solely to vertical lift and stabilization, and a single 50 cm pusher propeller positioned at the rear for forward propulsion. To meet the stringent marine environmental constraints, the airframe utilizes carbon fiber composite materials treated with marine-grade polyurethane. All internal avionics bays, payload mechanisms, and motor housings feature IP67-rated conformal sealing and sacrificial anode grounding to fulfill the corrosion-resistance requirement.
To achieve autonomous navigation and complex failsafe execution, the system employs a dual-layer, decentralized processing architecture:

- **Low-Level Control**: A robust flight control unit running PX4 firmware manages real-time attitude stabilization, three-loop autopilot gain scheduling, and the electromechanical mixing required during the critical transition phases between hover and cruise.
- **High-Level Cognition**: A companion edge computer (e.g., Raspberry Pi 5) running ROS2 serves as the primary decision making hub. This isolates computationally heavy tasks—such as computer vision, dynamic path planning, and sensor fusion—from the critical flight control loops.

Landing a VTOL with a 5 kg payload onto a platform moving at up to 2 m/s amidst 12 m/s crosswinds demands high-frequency perception. Standard GPS positioning is insufficient for this level of dynamic tracking and is highly vulnerable to jamming in contested zones. For this the drone utilizes a **downward facing optical flow sensor** (such as the PMW3901) feeding into a **YOLOv8-trained** neural network on the companion computer. This vision system actively detects and tracks the vessel's landing beacon or deck markers. Once the target is acquired, the ROS2 node calculates a predictive intercept trajectory based on the vessel's 2 m/s heading and the 12 m/s wind disturbance, sending continuous velocity setpoints via MAVROS to the flight controller to ensure a synchronized vertical descent.

Mission Flight Phases

- **Vertical Takeoff**: The four 25 cm lift rotors spool up to overcome the 26.9 kg MTOW, lifting the aircraft clear of ground obstacles or deck superstructure.
- **Outbound Transition**: The 50 cm pusher propeller engages. As forward airspeed increases and the 12-foot wings begin generating aerodynamic lift, power to the lift rotors is systematically tapered off until they lock into a low-drag orientation.
- **Forward Cruise**: The drone covers the 15 km outbound leg operating exclusively as a highly efficient fixed-wing aircraft, minimizing energy consumption.
- **Target Acquisition & Inbound Transition**: Upon reaching the destination coordinates, the lift rotors re-engage and the pusher propeller slows. The vision system initializes to locate the moving deck.
- **Dynamic Hover and Drop**: The aircraft enters a precision hover. The flight controller continuously rejects the 12 m/s wind disturbances while matching the vessel's 2 m/s movement. The payload is either safely deposited via landing or lowered via an automated winch.
- **Autonomous Return (Failsafe)**: Should ground-control communication sever at any point during the operation, the pre-programmed lost-link failsafe triggers. The VTOL climbs to a safe altitude, transitions back to fixed-wing cruise using its energy reserves, and navigates back to base relying on its VIO-generated odometry map.

3.3 Tools and Materials

Software & Simulation Tools

- **Flight Control Firmware:** PX4 Autopilot, handling low-level motor mixing, three-loop autopilot stabilization, and transition phase logic.
- **Edge Computing Framework:** ROS 2 (Robot Operating System 2), acting as the central nervous system for decentralized decision-making and high-level autonomy.
- **Computer Vision:** YOLOv8 neural network architecture utilized for dynamic target acquisition and precision landing on the moving vessel deck.
- **Simulation Environment:** Gazebo Sim, used for Software-In-The-Loop (SITL) validation of flight dynamics, 12 m/s wind disturbance rejection, and moving-platform intercepts prior to hardware deployment.
- **CAD and Structural Analysis:** Autodesk Fusion 360 to generate the required native files, STEP exports, and exploded drawings, alongside ANSYS for aerodynamic and structural load simulations.

Avionics & Electronic Hardware

- **Flight Controller:** Pixhawk 6X (FMUv6X architecture) for deterministic real-time control, equipped with isolated vibration damping and redundant IMUs.
- **Companion Computer:** Raspberry Pi 5 or NVIDIA Jetson Orin Nano to execute computationally heavy ROS 2 nodes, and YOLOv8 inferencing.
- **Primary Perception Sensor:** Optical flow sensor (e.g. PMW3901) for lightweight, low-compute drift estimation and odometry, reserving active computer vision strictly for the terminal precision landing on the moving vessel deck.
- **Telemetry & Failsafe Links:** Encrypted 900MHz/2.4GHz dual-band telemetry radios with built-in lost-link failsafe triggers.

Propulsion & Power System

- **Lift Actuators:** Four high-torque brushless DC motors paired with 25 cm car-

bon fiber propellers and high-amperage Electronic Speed Controllers (ESCs).

- **Pusher Actuator:** A high-efficiency brushless DC motor paired with the 50 cm pusher propeller for fixed-wing cruise.
- **Lift Power Supply:** 12S (44.4V) 10,000 mAh LiPo battery pack designed for high instantaneous power draw (up to 4000 W).
- **Cruise & Avionics Power Supply:** Custom 12S4P Li-ion pack utilizing 21700-size cells (e.g., Molicel P42A) for high energy density during the 30 km round trip.
- **Delivery Mechanism:** Automated electromechanical winch or actuated payload bay door tailored to safely release the 5 kg cargo.

Structural Materials & Marine Resources

- **Airframe Skeleton:** Carbon fiber composite tubing and molded aerodynamic shells to support the 12-foot wingspan while minimizing weight.
- **Corrosion Protection:** Marine-grade polyurethane external coatings, IP67-rated conformal coatings for all exposed PCBs, sealed motor stators, and 316 stainless steel hardware to survive the saltwater environment.
- **Prototyping & Testing Instruments:** Digital multimeters, digital oscilloscopes, and a motor thrust stand for analyzing specific thrust curves and optimizing the QuadPlane power budget.

4 Implementation

4.1 Design and Development

Aerodynamic constraints and Optimization

The preliminary sizing is a constrained optimization: it finds the lightest aircraft that still completes the 15 km ship-to-shore mission in a 12 m/s wind. The design variables are wing area S, aspect ratio AR, cruise speed V and lift rotor diameter. For each combination, the code closes the mass budget by iteration, because every extra kilogram of structure needs more battery and more lift power. Each extra kilogram of component mass adds about 2.6 kg to MTOW. The result is a maximum takeoff weight and a power, energy and mass breakdown for that combination.

The **feasible region** is represented using constraint diagrams and design-space maps. The final design is selected at the lowest-weight feasible point while satisfying the specified performance and safety margins. The selected configuration, consisting of a NACA 4415 airfoil, wing area of $S = 1.48$ m^2, aspect ratio of $AR = 9.0$, cruise velocity of $V = 24$ m/s, and maximum take-off weight of $MTOW = 26.9$ kg, satisfies all the specified constraints. The gust stall margin is the most restrictive constraint, with a margin of 0%, thereby limiting the potential for further reduction in wing area.

The figure 3 shows the variation of lift-to-drag ratio (L/D) with lift coefficient (C_L). The maximum L/D of 12.0 occurs at $C_L \approx 0.94$, indicating the best aerodynamic efficiency. The selected cruise condition is $C_L \approx 0.50$, with $L/D \approx 10.1$.

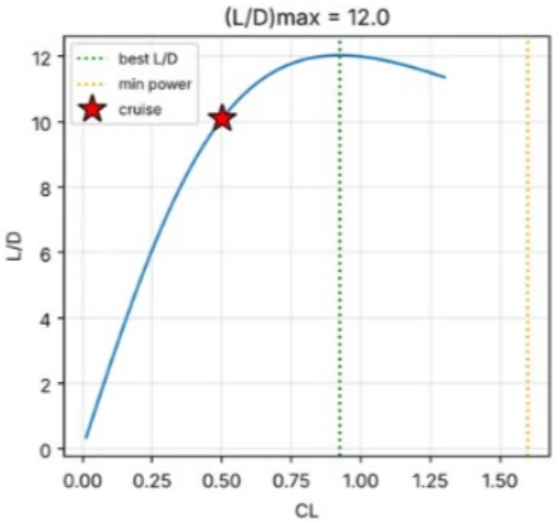


Figure 3: L/D ratio vs C_l

The lift curve illustrates 4 the variation of lift

coefficient (C_L) with angle of attack (α). The lift-curve slope is 4.90 rad^{-1}, with a maximum lift coefficient of 1.30 in the clean configuration and 1.55 with flaps deployed. The increased lift capability with flaps improves low-speed performance.

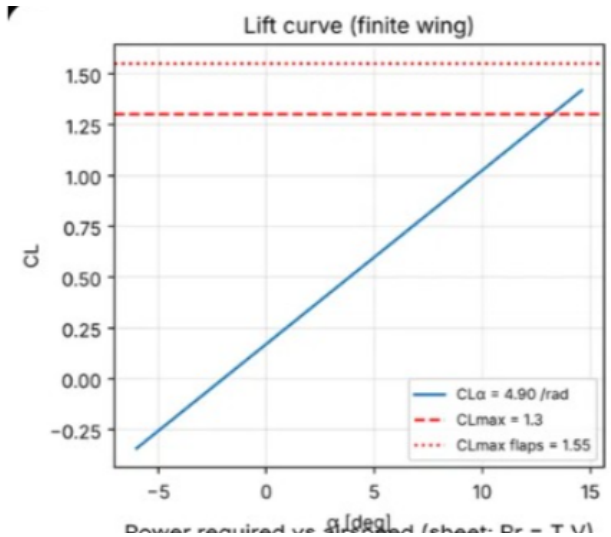


Figure 4: C_l vs alpha(deg)

The constraint diagram 5 presents the required thrust-to-weight ratio (T/W) against wing loading (W/S) for cruise, climb, and transition acceleration requirements. The shaded region represents the feasible design space, while the red star indicates the selected design point at $W/S \approx 179$ N/m^2 and $T/W \approx 0.11$. The selected point satisfies the cruise requirement but requires further verification against climb and transition constraints.

The figure 6 shows the required thrust-to-weight ratio (T/W) as a function of angle of attack (α) for climb angles of 0°, 5°, and 10°. Higher climb angles require greater thrust, while the minimum thrust requirement for level flight occurs near $\alpha \approx 8.5°$. The vertical line at $\alpha \approx 4°$ indicates the selected cruise condition at 24 m/s.

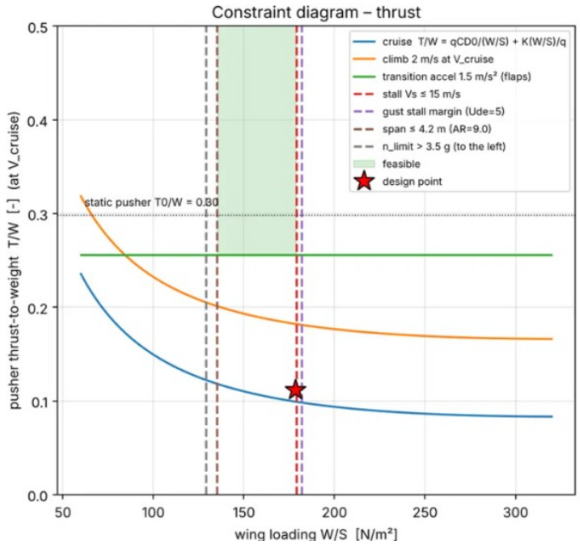


Figure 5: Pusher Thrust to weight ratio vs wing loading

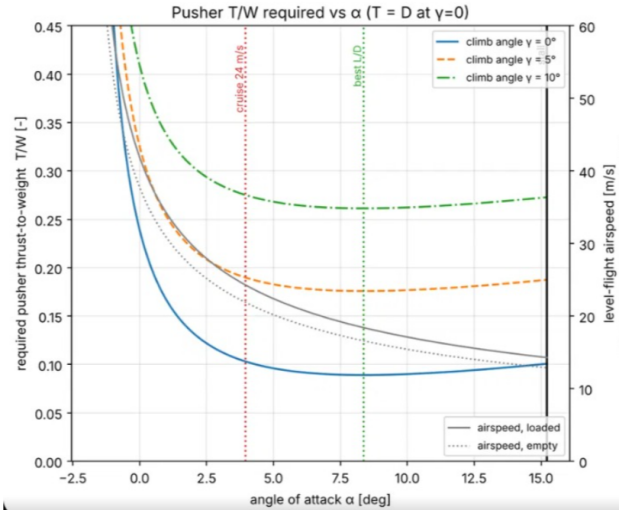


Figure 6: Pusher Thrust to weight ratio vs alpha

Airfoil Selection

The NACA 4415 airfoil was selected for the present wing design because of its potential to provide a good balance between lift generation, aerodynamic efficiency and structural practicality. The airfoil has a maximum camber of 4% of the chord at 40% chord and a maximum thickness of 15% of the chord, making it a moderately cambered and relatively thick profile. Previous studies have investigated its suitability for low-speed UAV applications. Saeed et al. (2025) compared NACA 4415 with NACA 2412 and NACA 0012 using CFD and XFOIL, reporting favourable lift-to-drag performance for NACA 4415 under their investigated conditions. Its relatively large thickness also offers additional space for structural components within the wing. In the present study, NACA 4415 was chosen for further evaluation because its geometry and aerodynamic characteristics were considered suitable for the intended operating conditions.

The preliminary analysis using Foil.tools gave a lift coefficient of approximately 0.52 and a drag coefficient of 0.0074 at an angle of attack of 1.5°. These results supported its inclusion in the final design selection, with its suitability assessed against the required lift, drag and pitching-moment characteristics.

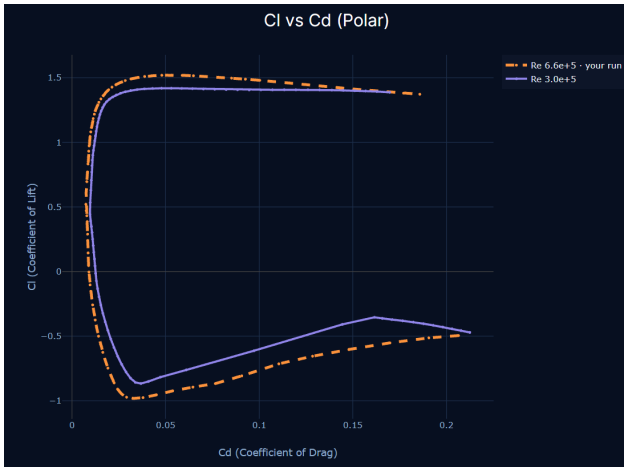


Figure 7: Cl:Cd for naca4415

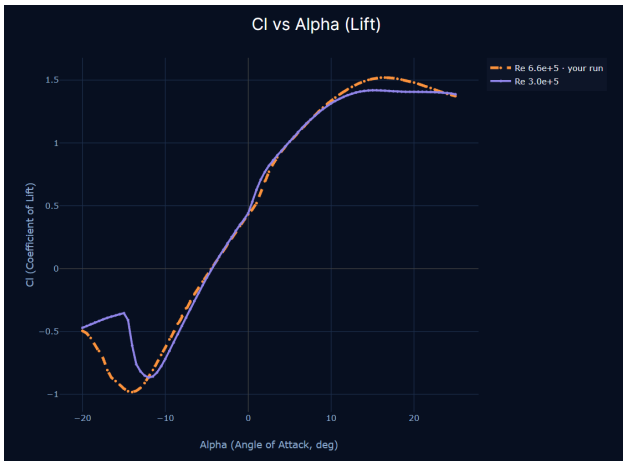


Figure 8: C_l vs alphs for naca 4415

4.2 Power Budget

A 12S 10000mAh battery meets our lift system's demands by operating at a high enough voltage to keep electrical current manageable, while storing enough total capacity to clear our flight phase energy budget with a healthy reserve.

Table 1: Aerodynamic Performance Parameters

Symbol	Value
V_{cruise}	24 m/s
W/S	179 N/m^2
$(L/D)_{\max}$	12.0
$C_{L,(L/D)_{\max}}$	0.94
$C_{L,\text{cruise}}$	0.50
$(L/D)_{\text{cruise}}$	10.1
V_{stall}	15 m/s
$C_{L,\max}$	1.30
$C_{L,\max,\text{flaps}}$	1.55
C_{L_α}	4.90 rad^{-1}
C_{D0}	0.0384
e	0.78
K	0.0450
Re_{MAC}	6.62×10^5

Table 2: Geometric Parameters of the Aircraft

Symbol	Value
b	3.704 m
S	1.495 m^2
AR	9.0
c_{root}	0.515 m
c_{tip}	0.293 m
$\bar{c}$	0.404 m
MAC	0.414 m
θ	83.2°
h	0.300 m
b_{local}	0.300 m
l	0.950 m
t/c	0.15
t_{root}	0.07725 m
t_{tip}	0.04395 m
L_f	1.900 m
W_f	0.400 m
H_f	0.400 m
V_{avail}	0.0855 m^3
D_{rotor}	28 in

1. Handling the 4000 W Instantaneous Power Draw

Our energy budget states that the takeoff hover, ship hover, and winch drop phases require an instantaneous power draw of ∼4000 W. To determine how much electrical current the battery must deliver to achieve this, we use the electrical power equation:

$$I = \frac{P}{V}$$

$$I = \frac{4000 \text{ W}}{44.4 \text{ V}} \approx 90.1 \text{ A}$$

Drawing 90.1 amps is a heavy load, which is why the battery's discharge rating (C-rate) is critical. The C-rate required to supply this current from a 10,000 mAh (10 Ah) pack is calculated as:

$$C_{\text{rate}} = \frac{I_{\text{draw}}}{\text{Capacity}} = \frac{90.1 \text{ A}}{10 \text{ Ah}} \approx 9\text{C}$$

Standard lithium polymer (LiPo) drone batteries are commonly rated for 25C, 50C, or even higher continuous discharge rates. A 9C peak discharge requirement is comfortably low for a quality LiPo, meaning the battery will deliver the 4000 W without experiencing excessive voltage sag, thermal throttling, or component damage.

2. Satisfying the Mission Energy Capacity

The total energy capacity of a battery in Watt-hours (Wh) dictates how long it can sustain flight. It is calculated by multiplying the nominal voltage by the amp-hour capacity:

$$E = 44.4 \text{ V} \times 10 \text{ Ah} = 444 \text{ Wh}$$

According to your power budget, the vertical flight phases combined (Takeoff Hover 52 Wh, Ship Hover + Winch Drop 135 Wh, 4x Flight Transitions 27 Wh, Landing Hover 57 Wh) consume a base total of 271 Wh.

The 444 Wh pack provides exactly 173 Wh of excess capacity for the lift system alone. This surplus represents a ∼39% safety margin above the base requirement, comfortably satisfying the mandatory 30% reserve for abort scenarios, wind rejection, and dynamic deck tracking outlined in your problem statement.

Table 3: Subsystem power budget (MTOW = 26.5 kg)

Subsystem	**Cont. / Peak**
4× Lift motors	4.6 kW / 8.6 kW
Pusher (⌀ 500)	960 W / 1.6 kW
Avionics	40 W / 60 W
Pack Draw	**5.6kW / 10.26 kW**

Table 4: Bill of Materials and Estimated Cost Breakdown

Subsystem / Component	**Qty**	**Cost (INR)**
Avionics & Navigation		
Holybro Pixhawk 6X Flight Controller	1	31,079
Raspberry Pi 5 (8GB) Edge Companion	1	7,500
Holybro PMW3901 Optical Flow Sensor	1	3,149
Downward-Facing CSI/USB Camera Module	1	2,500
900 MHz Encrypted Telemetry Transceiver Set	1	8,000
Subtotal		*52,228*
Power Systems		
Tattu 6S 10,000 mAh LiPo Packs (12S Lift)	2	34,328
Molicel P42A 21700 Li-ion Cells (12S4P Cruise)	48	24,480
Custom BMS, Nickel Strips & Pack Enclosure	1	3,500
Subtotal		*62,308*
Propulsion Systems		
Heavy-Lift BLDC Motors (600–800 kV)	4	32,000
High-Efficiency Pusher BLDC Cruise Motor	1	6,500
12S ESCs (4×60A Lift, 1×80A Pusher)	5	17,500
Carbon Fiber Propellers (4×25 cm, 1×50 cm)	5	6,000
Subtotal		*62,000*
Airframe & Marine Protection		
Composite Airframe (Carbon Booms, Foam Wings)	1	45,000
Marine Coatings (IP67 Conformal, 316 SS Hardware)	1	6,000
Subtotal		*51,000*
Hardware Subtotal		**2,27,536**
Contingency Buffer (10%)		23,303
Total Estimated Budget		**2,50,839**

5 Results and Discussion

5.1 Simulation

Development of the simulation environment using ROS 2, Gazebo Sim, and PX4.

- Simulation Environment: Developed a custom Gazebo world (varuna_mission.sdf) featuring a large ocean, an island with

Table 5: Estimated Aircraft Mass Breakdown

Group	Component	Mass (kg)
Fixed	Payload	5.00
	Avionics	0.90
	Wiring	0.50
	Fuselage	1.80
	Subtotal	*8.20*
Airframe	Wing	2.96
	Tail surfaces	0.72
	Booms	1.34
	Corrosion protection	0.20
	Subtotal	*5.22*
Propulsion and power	Lift motors, ESCs,	4.02
	Pusher system	0.71
	Battery	7.16
	Subtotal	*11.89*
	Total	**25.31**

vegetation, a supply vessel, and a moving landing platform to represent the maritime UAV mission environment.

- - World Configuration: Configured the simulation environment, including physics parameters, gravity, magnetic field, spherical coordinates, and simulation update rate, while referring to the stock PX4 Gazebo environment.
- Wind Control: Developed a Python-based GUI to control wind speed and direction during simulation. Configured the environment to support wind speeds up to 12 m/s and tested direct wind updates through Gazebo topics.
- Vessel and Landing Platform Control: Developed a Python GUI to adjust the vessel's position and control the landing platform's movement. Implemented position updates through Gazebo's set_pose service, with configurable platform speeds from 0.1 to 2 m/s.
- PX4–Gazebo Integration: Worked on integrating the custom simulation environment with PX4, investigated sensor communication and configuration and established a working simulation setup for UAV testing.

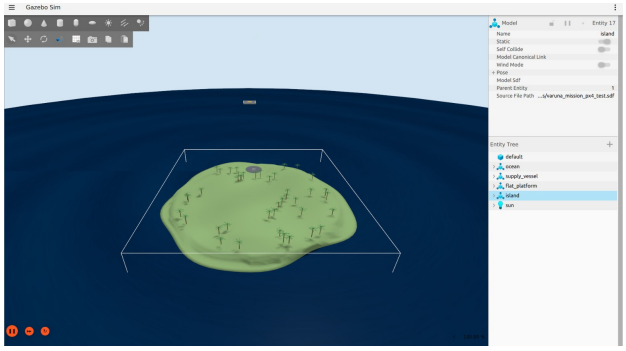


Figure 9: Island environment in Gazebo

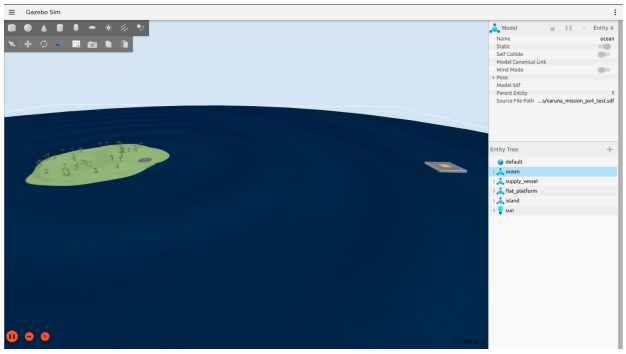


Figure 10: Island with vessel environment in Gazebo

- Custom UAV Model Preparation: Integrated the custom thunderb_vtol model into the simulation workflow and worked on its compatibility with PX4 SITL and the Varuna world.
- Hover Testing and Verification: Successfully demonstrated takeoff and approximately 10 seconds of stable hover by the custom VTOL UAV in the Varuna simulation environment.
- Debugging and Version Control: Troubleshot Gazebo configuration, resource paths, simulation partitions, and model-control issues. Used Git and GitHub to maintain and upload the simulation work to the team's project repository.

Tools and Technologies Used

Ubuntu 22.04, ROS 2 Humble, Gazebo Sim 8.15.0, PX4 Autopilot v1.17.0, Python, Tkinter, SDF, Gazebo CLI, Git, and GitHub.

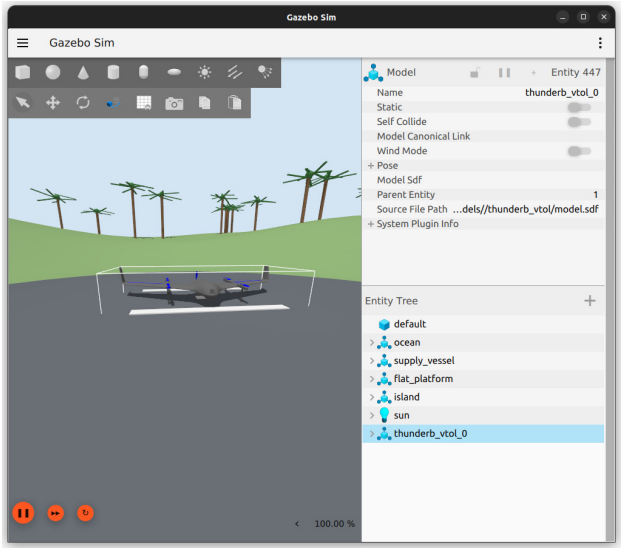


Figure 11: VTOL simulation in the custom world , thunderb_vtol_0 is taken for simulation testing only

5.2 CAD Models

The current CAD model (fig. 12) establishes a robust, twin-boom airframe designed to balance structural rigidity with aerodynamic efficiency. The architecture centers around a high-volume fuselage pod for optimal weight distribution and an inverted-U tail configuration to keep control surfaces clear of central propwash. To maximize cruising endurance and payload capacity during long range operations, the aircraft utilizes tapered main wings built around a high-lift NACA 4415 airfoil profile.

Subsequent development phases will integrate the propulsion and control subsystems required for full hybrid VTOL capability. Four vertical-lift motors and propellers will be mounted along the longitudinal booms to govern stable hovering and vertical takeoff, while a primary rear pusher motor will drive forward cruise flight. The central fuselage pod will securely house the primary avionics suite, alongside a downward-facing optical flow sensor to enable precision hovering and target-oriented landing capabilities in environments where GPS reliability is limited.

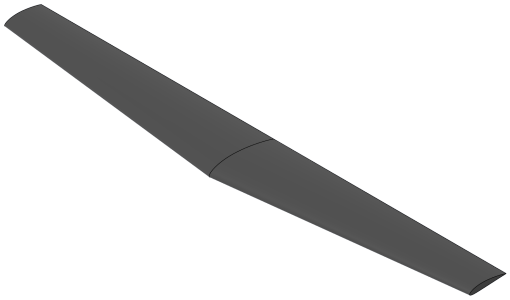


Figure 13: NACA 4415 airfoil wings

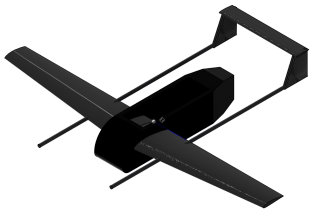


Figure 12: Assembly

6 Conclusion & Future Scope

6.1 Conclusion

The project presents the design and simulation of a hybrid VTOL UAV for ship-to-shore and island supply missions. The work includes preliminary aerodynamic analysis, airframe design, power budgeting, and the development of a custom maritime simulation environment using ROS 2, Gazebo Sim, and PX4. The simulation environment incorporates an ocean, an island, a supply vessel, a moving landing platform, and configurable wind conditions. Successful take-off and approximately 10 seconds of stable hovering by the custom VTOL model demonstrated the feasibility of the initial simulation setup. Overall, the project establishes a foundation for further development and testing of an autonomous UAV capable of delivering payloads in challenging maritime environments.

6.2 Future Scope

Future work will focus on improving the UAV model and validating its performance under realistic operating conditions. The simulation can be extended to include autonomous navigation, waypoint-based mission planning, precise landing on moving platforms, and payload delivery. Further testing under wind speeds of up to 12 m/s and platform velocities of up to 2 m/s will help evaluate flight stability and control performance. The next stages can also include integrating computer vision and additional sensors, implementing communication-loss failsafes, and validating the transition between vertical flight and fixed-wing cruise. Finally, hardware testing and experimental validation will be required to assess the aircraft's endurance, payload capacity, and suitability for real-world maritime resupply missions. hackathon

AI disclosure

AI tools were used as development and troubleshooting assistants to help structure simulation configuration, develop and refine Python GUI scripts, identify configuration errors, interpret terminal outputs, and suggest debugging approaches. The suggested changes were reviewed and tested in the local simulation environment. AI assistance did not replace practical testing, and no unverified flight-performance results are claimed.

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References

[1] Haowei Gu, Ximin Lyu, Zexiang Li, Shaojie Shen, and Fu Zhang. Development and experimental verification of a hybrid vertical take-off and landing (vtol) unmanned aerial vehicle(uav). In *2017 International Conference on Unmanned Aircraft Systems (ICUAS)*, pages 160–169, 2017.

[2] M. Rehan, F. Akram, A. Shahzad, T.A. Shams, and Q. Ali. Vertical take-off and landing hybrid unmanned aerial vehicles: An overview. *The Aeronautical Journal*, 126(1306):2017–2057, 2022.